Dennis G. Pezan

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Education

United States Military Academy at West Point — West Point, NY
B.S., Operations Research (Finance Specialization), Minor in Artificial Intelligence — Expected May 2029
- CQPA: 3.571 | Dean's List, Fall 2025 and Spring 2026
- Research Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply (Advisor: LTC Matthew Webb)

Research Experience

Contested Logistics / Network Interdiction Research – D/Math, USMA — West Point, NY
Undergraduate Researcher (Advisor: LTC Matthew Webb) — Jul 2026 – Present
- Extend LTC Webb's doctoral research on reinforcement learning for maximum-flow network interdiction to a 57-node, 79-edge open-source model of Taiwan's crude oil import network with sourced capacities and seven costed adversary actions anchored on historical events
- Connected the network to a reinforcement-learning interdiction environment, verified it against an independent max-flow calculation, and fixed a targeting-filter issue that could hide the Strait of Hormuz
- Score each adversary plan by the expected share of Taiwan's crude demand still delivered after rerouting; the agent must beat random plans (about 90%) and a greedy baseline (70%) and approach an exact solver across six sensitivity cases

Reinforcement Learning for Contested Resupply – TDAC, U.S. Army — Fort Lee, VA
Research Intern, Modeling & Analysis Division (AIAD; Mentor: MAJ Erica Mikelson) — Jul 2026
- Redesigned the reward function for a proximal policy optimization (PPO) reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's stochastic sustainment simulation; trained and evaluated the agent in Python (PyTorch)
- Built two diagnostic tools that exposed four reward flaws, including an agent that outscored every baseline while starving the priority unit; the retrained agent more than tripled the share of requests filled and beat a hand-coded priority-aware baseline across 101 random seeds
- Presented findings to Branch Chief LTC Ryan Miller at Fort Lee and, by call, to TDAC's AI and reinforcement-learning researchers (LTC Joseph Pedersen) at the Naval Postgraduate School in Monterey, CA

Tree-Based Methods in Machine Learning – D/Math, USMA — West Point, NY
Independent Study in Machine Learning (MA289), Advisor: LTC Patrick Kuiper — Oct 2025 – May 2026
- Designed and completed an independent study of decision trees, bagging, and random forests, and presented the findings to officers
- Built an interactive random forest research site with a live classifier demonstration and hyperparameter explorer

Projects

Tree-Based Methods Research Site – dennis-pezan.com/ma289.html — 2026
- Trained and evaluated decision trees, bagging, and random forests on the Titanic dataset in Python (scikit-learn) on an identical 80/20 split
- Computed pairwise tree correlation, overfitting gaps, and feature importance to show how ensemble decorrelation reduces variance; deployed via GitHub and Vercel

CFT Packing List – dennis-pezan.com/cft.html — 2026
- JavaScript web tool that helps cadets organize and track gear for Cadet Field Training; saves progress in the browser and deploys on Vercel from GitHub

Reward Engineering Case Study – dennis-pezan.com/rl-reward.html — 2026
- Public write-up diagnosing and correcting reward hacking in a simulated shipping network and raising requests met from 24% to 81%

Mounger Writing Center Website Redesign – dennis-pezan.com/mwc-redesign.html — Aug 2026 – Present
- Independent redesign concept for West Point's writing center site, built in Next.js, TypeScript, and Tailwind CSS with a custom WebGL ink shader and editable content files for staff

Professional Development

Sea to Shining Sea Domestic AIAD — California and Colorado
Summer AIAD on Defense Technology and Innovation (separate from the academic-year fellowship) — Jun 2026
- Held in-depth discussions with leaders at Palantir, Stanford's Gordian Knot Center, OpenAI, Apple, and Meta Reality Labs on how defense technology is developed, procured, and fielded
- Topics: startup barriers to Army adoption, agentic AI and large language models for defense, supply-chain risk, and AI regulation
- Observed how the 4th Infantry Division employs and gives feedback on these systems under Next Generation Command and Control; visited U.S. Northern Command, NORAD, and the Cheyenne Mountain Complex

Honors & Awards

Dean's List, United States Military Academy — Fall 2025, Spring 2026
Stokes Writing Fellowship, United States Military Academy — 2026–Present
- Endowed fellowship funding research travel, expert mentorship, and publication of original research

Norwegian Foot March Badge, Norwegian Armed Forces — 2026
Sea to Shining Sea Fellowship, United States Military Academy — 2026–Present
- Academic-year fellowship engaging cadets with senior leaders in national security, government, finance, and technology; separate from the summer Domestic AIAD

Leadership & Service

Center for Enhanced Performance, USMA — West Point, NY
Cadet-in-Charge, MA103 Data-Driven Modeling Tutoring — Aug 2026 – Present
- Lead the full tutor staff for MA103, a core mathematics course with 1,010 cadets enrolled; direct tutoring coverage for every major graded event
- Built a study site (dennis-pezan.com/ma103-tutoring.html) with two full-length practice exams, a mirror of a graded gate, and a 59-drill reference card, each with an answer key

Mounger Writing Center, USMA — West Point, NY
Writing Consultant — Aug 2026 – Present
- Selected as a Stokes Writing Fellow to coach cadets on strengthening the rhetoric and argument of their writing

CLD-STEM Outreach Program, USMA — West Point, NY
Member — Oct 2025 – Present
- Bring hands-on STEM projects to students, reaching 800+ students across the Chicagoland and Greater Los Angeles areas

Cadet Ambassador Program, USMA — West Point, NY
Cadet Ambassador — Jan 2026 – Present
- Represent West Point in outreach to prospective cadets and help foster a welcoming environment for future applicants

Finance Club, USMA — West Point, NY
Member — Oct 2025 – Present
- Engage with industry executives from companies including Palantir, Uber, and Meta on leadership in business, business ethics, and personal finance

Recycle & Reboot — Plainfield, IL
Volunteer Senior Publicist — May 2024 – Aug 2024
- Led outreach to secure donated technology; helped refurbish and distribute 100+ computers to underserved communities

Illinois Premier Boys State — Illinois
Candidate — Jun 2024
- Elected Alderman and Township Supervisor; advanced to the Lieutenant Governor primary through competitive public speaking

Professional Experience

McDonald's — Plainfield, IL
Department Manager — Sep 2022 – Jul 2024
- Promoted at age 16 as the region's youngest department manager; sustained 90%+ customer satisfaction scores
- Scheduled 80+ employees weekly with 95% on-time accuracy; oversaw $20,000+ in weekly supply procurement

GrayStone AI Solutions — Remote
Intern — Jun 2024 – Jul 2024
- Built Python and Excel tools to support financial planning and analysis workflows

Skills

Technical: Python (PyTorch, scikit-learn), SQL, Java, JavaScript/TypeScript (Next.js) | **Methods:** reinforcement learning, network optimization, machine learning
Certifications: JPMorgan Chase Software Engineering Job Simulation (2026); Microsoft Fabric Data Architecture (2026)
Languages: Spanish (fluent) | **Clearance:** Secret